Paleomagnetism of the Fond du Lac Formation and the Eileen and Middle River sections with implications for Keweenawan tectonics and the Grenville problem

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Overlying the Keweenawan lavas of northern Wisconsin and Michigan is a thick sequence of terrestrial sandstone, shale, and siltstone that has undergone some tectonic deformation associated with movement along thrust faults and the development of the Lake Superior syncline. Thermal and alternating field demagnetization, chemical leaching, and multivector analysis using orthogonal projections reveal a trivector structure of the natural remanent magnetization (NRM) of the Fond du Lac Formation and Middle River section (Amnicon and Orienta Formations), and a bivector structure of the NRM of the Eileen section (Eileen Formation). The components may be classified by their physical properties as revealed by demagnetization. A population of high blocking temperature components, K_1, is found in all three sections and gives poles as follows: Fond du Lac, 16°N, 160°E; Middle River, 25°N, 148°E; Eileen, 20°N, 156°E after structural correction is applied. A population of intermediate blocking temperature components, K_2, is post-tectonic and found only in the Fond du Lac Formation and Middle River section. Poles calculated from K_2 fall among the Grenville type poles (Fond du Lac, 9°S, 145°E; Middle River, 24°S, 162°E). A third population of components, K_3, has low blocking temperature and coercivity and is isolated only by chemical leaching. K_3 has steep positive inclination, northern declination, and is post-tectonic. It is interpreted as a recent magnetization.

Any interpretation of the path of apparent polar wander for North America must accommodate the sequence of magnetization K_1 to K_2. The timing of tectonism in the Keweenawan basin is bracketed by the times of acquisition of K_1 and K_2. These results reconfirm some recent interpretations that include Grenville poles on the polar wander track of interior Laurentia.

Reposant sur les laves du Keweenavien dans le nord du Michigan et du Wisconsin se trouve une épaisse séquence de grès, de shales et de siltstones terrestres qui ont subi une certaine déformation tectonique associée au mouvement le long de failles de chevauchement et au développement du synclinal du Lac Supérieur. La désaimantation thermique et dans un champ alternatif, le lessivage chimique et l'analyse multivectorielle en utilisant des projections orthogonales révèlent une structure tri-vectorielle de l'ANR (aimantation naturelle rémanente) de la formation de Fond du Lac et de la coupe de Middle River (formations d'Amnicon et d'Orienta), et une structure bi-vectorielle de l'ANR pour la coupe d'Eileen (formation d'Eileen). On peut classifier les composantes par leurs propriétés physiques comme les révèle la désaimantation. Une population de composantes à haute température de blocage, K_1, s'observe dans les trois coupes et donne les pôles suivants: Fond du Lac, 16°N, 160°E; Middle River, 25°N, 148°E; Eileen, 20°N, 156°E après application de la correction structurale. Une population de composantes à températures de blocage intermédiaires, K_2, est post-tectonique et se retrouve seulement dans la formation de Fond du Lac et la coupe de Middle River. Les pôles calculés à partir de K_2 tombent parmi les pôles typiques du Grenville (Fond du Lac, 9°S, 145°E; Middle River, 24°S, 162°E). Une troisième population de composantes, K_3, a une coercivité et une température de blocage faibles et elle ne peut être isolée que par un lessivage chimique. K_3 a une inclinaison positive abrupte, une déclinaison vers le nord et elle est post-tectonique. On l'interprète comme une aimantation récente.

Toute interprétation de la trajectoire du cheminement polaire apparent pour l'Amérique du Nord doit accomoder la séquence d'aimantation de K_1 à K_2. La chronologie du tectonisme dans le bassin du Keweenavien est limitée par la fourchette des temps d'acquisition de K_1 et de K_2. Ces résultats reconfirment quelques interprétations récentes qui placent les pôles du Grenville sur la trace du cheminement polaire pour l'intérieur de la Laurentie.

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Introduction

The meaning of the paleomagnetic signature of the Grenville Province of Canada has been a matter of discussion in recent years. Interpretations range from those that consider the Grenville paleomagnetic poles an integral part of the North American polar wander track (Morris and Roy 1977; McWilliams and Dunlop 1978; Berger *et al.* 1979) to those that consider the possibility of movement of the Grenville Province with respect to Interior Laurentia after the magnetization of the Grenville rocks (Irving *et al.* 1974; Ueno *et al.* 1975). The crux of the matter is the apparent uniqueness of the Grenville poles in the North American paleomagnetic record.

Presented here are paleomagnetic results from the Upper Keweenawan sedimentary rocks of Wisconsin and Minnesota. These rocks have undergone up to three phases of magnetization but the components of the natural remanent magnetization (NRM) are resolved by

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TABLE 1. Stratigraphy of the Oronto and Bayfield Groups

According to Thwaites (1912)		According to Tyler *et al.* (1940)	
Bayfield Group		Bayfield Group	
Chequamegon sandstone	305 m	Chequamegon sandstone	305 m
Devil's Island sandstone	92	Devil's Island sandstone	92
Orienta sandstone	915	Orienta sandstone	915
Oronto Group		Oronto Group	
Amnicon sandstone	1525		
Eileen sandstone	610		
Freda sandstone	3660	Freda sandstone	3660
Nonesuch shale	107	Nonesuch shale	107
Outer conglomerate	366	Outer conglomerate	366
	7580 m		5445 m

thermal and alternating field demagnetization and chemical leaching. The absolute ages of magnetization are uncertain because they are recorded in Precambrian sedimentary rocks. However, the simple style of deformation allows unambiguous assignment of pre- and post-tectonic ages to the components of the NRM.

It will be demonstrated that the paleomagnetic signature of the Upper Keweenawan rocks investigated in this study is very similar to that reported from certain rocks in the Grenville Province. The sequence of magnetization is consistent with those interpretations that include the Grenville poles on the polar wander track of Interior Laurentia. Comparing poles calculated from pre- and post-tectonic magnetizations in the Keweenawan to dated poles from igneous and metamorphic rock units yields an estimate of the time of deformation in the Keweenawan Basin.

Geology of the Upper Keweenawan

Cropping out along the south shore of Lake Superior and various natural exposures along streams and rivers in northern Wisconsin, Michigan, and east-central Minnesota is a thick (5000 m) accumulation of terrestrial conglomerates, sandstones, siltstones, and shales generally assigned to the Upper Keweenawan. The stratigraphic position of these rocks is poorly determined because of the complete absence of a fossil record and limited exposure due to the cover of Pleistocene drift. It is known that these rocks overlie the ca. 1100 Ma Keweenawan basalts and, in the Upper Peninsula of Michigan, are overlain by Upper Cambrian sediments. These rocks have undergone structural deformation, manifested by the warping of Keweenawan and underlying rocks into a broad syncline now occupied by Lake Superior, and substantial movement along several thrust faults (Thwaites 1912; Halls 1966; White 1966*a*,*b*). These structures do not penetrate the Upper Cambrian of northern Michigan.

The Upper Keweenawan sediments in Wisconsin have been divided into two groups (Thwaites 1912), the Oronto Group and the overlying Bayfield Group. The Oronto Group has a basal conglomerate in some localities (Copper Harbor conglomerate and Outer conglomerate), which overlies the Middle Keweenawan volcanics in a nonconformable contact. The conglomerate grades upward into a series of shales and immature arkosic sandstones. The Bayfield Group is subarkosic to arkosic in the lowest portion of the section but grades upward into a relatively pure quartzose sandstone. Several subdivisions of these groups have been presented, two of which (Thwaites 1912; Tyler *et al.* 1940) are summarized in Table 1.

The stratigraphy and petrology of the Fond du Lac Formation have been summarized by Morey (1967) and much of the following description is from that work. The Fond du Lac Formation consists of subarkosic and arkosic shales, siltstones, and sandstones with a basal conglomerate, probably representing a wedge of clastic material deposited in an oxidizing deltaic environment. These rocks crop out in a limited area along the St. Louis River and a small tributary, Mission Creek, near Duluth, Minnesota (Fig. 1). The Fond du Lac exceeds 244 m in thickness, but only 90 m is now exposed. The red color is due to hematite, which occurs as a coating on the surfaces of sand- and silt-sized quartz and feldspar grains or occupying the interstices between the grains. Hematite was probably formed in place, sometime after deposition and prior to complete lithification. A further source of hematite may be the alteration of detrital iron-bearing silicate minerals such as pyroxene, hornblende, and biotite.

The correlation and absolute age of the Fond du Lac Formation have long been problems. It overlies with profound angular unconformity the Animikean Thompson Formation, deformed during the Penokean Orogeny ca. 1.7 GA. It is probable that the Fond du Lac overlies the Middle Keweenawan basalts in the area. The Fond du Lac Formation has been correlated with the Oronto–

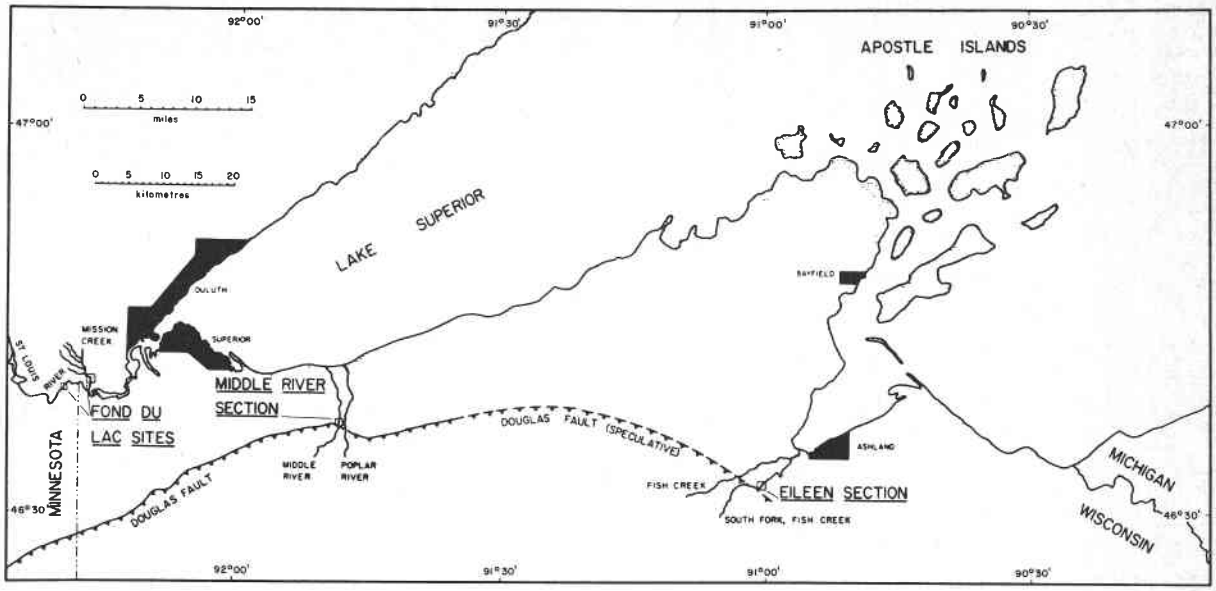


FIG. 1. Map of southwest Lake Superior showing sampling localities and the Douglas Fault (after Thwaites 1912; Halls 1966).

Bayfield boundary by Thwaites (1912) and the Lower Bayfield by Tyler *et al.* (1940) (see Table 1).

Approximately 915 m of subarkosic red sandstone, siltstone, and shale is exposed in a vertically dipping to overturned section on the Middle River of Wisconsin (Fig. 1). The tilting of the beds is interpreted to have resulted from drag along the Douglas Fault. The Douglas Fault is clearly exposed at the southern end of the section where basalt is in fault contact with the sedimentary rock. Thwaites (1912) and Tyler *et al.* (1940) assign the lowest 115 m to the Oronto Group on the basis of petrology and heavy mineral content, suggesting that the contact between the Bayfield and Oronto Groups is conformable and gradational.

The Eileen Formation was defined by Thwaites (1912) from a nearly continuous exposure of vertically dipping sandstones and shales cropping out along the south fork of Fish Creek near Ashland, Wisconsin (Fig. 1). This section consists of varigated (red to white mottled) cross-bedded, horizontally jointed, subarkosic sandstone, siltstones, and shales. Petrographically, these rocks are very similar to those of the Bayfield Group. Thwaites assigned the Eileen section to the Oronto Group because of the occurrence of a section assigned to the Oronto Group in an up-dip direction along Fish Creek, assuming these exposures are a stratigraphically continuous section. Tyler *et al.* (1940) suggested that the petrology and heavy mineral content of the Eileen section show these rocks belong to the Bayfield Group. They offered a number of hypothetical structures that could have brought the Eileen section to an apparent down-dip position with respect to the Oronto Group rocks. Because the classification of the rocks along the south fork of Fish Creek is questionable, they will be referred to as the Eileen section rather than the Eileen Formation.

Paleomagnetic sampling

Samples were collected with a portable coring apparatus and oriented with a Brunton compass and clinometer.

A preliminary study of the Fond du Lac Formation (Watts and Van der Voo 1977) revealed that thermal demagnetization of the coarse- to medium-grained rocks resulted in nonlinear, discontinuous demagnetization trajectories and that demagnetization of the fine-grained samples often gave well-defined characteristic and secondary directions. A second, intense sampling was conducted in the Keweenawan Basin in an effort to collect the finest grained, nonfriable rocks.

Seventy-four cores were collected from 12 sites in the siltstone and shale beds of the Fond du Lac Formation along the banks of the St. Louis River and Mission Creek. Twenty-six cores were collected from 420 m of the vertical to overturned Middle River section. The sampling was concentrated in the siltstones and fine sandstones. Samples were gathered from rocks assigned to the Oronto and Bayfield Groups. The up-section direction was determined from cross-beds. Sixteen cores were collected from 470 m of the Eileen section with one or more cores taken from each siltstone and fine sandstone layer encountered.

The purpose in sampling the tilted sections was to determine the relationship between the timing of magnetization and structural deformation in the Keweenawan Basin. There is little evidence to show that the sections

are of the same age so this is not, strictly speaking, the fold test of Graham (1949). It is assumed that the path of apparent polar wander for Upper Precambrian North America is sufficiently well documented that a determination of pre- or post-tectonic age of magnetization can be inferred by comparison of results to well-documented poles from Laurentia for the period 1100–650 Ma.

Laboratory analysis

Standard cylinders were cut from the cores and measured with a ScT cryogenic magnetometer and a SSM-1A Schonstedt spinner magnetometer. Alternating field and thermal demagnetization were carried out using Schonstedt GSD-1 and TSD-1 units, respectively. Chemical leaching was done by cutting three parallel slots perpendicular to the bedding in the specimens, and placing the specimens in 8 *N* HCl in a reduced field (approximately 10^{-4} mT) for periods between 10 and 20 h. After leaching, the specimens were washed and stored in distilled water in a reduced field for periods of 5 h, and dried in a reduced field. After 80 cumulative hours of leaching in 8 *N* HCl, the acid concentration was increased to 10 *N*, and demagnetization continued until it was evident that further treatment was not removing significant amounts of the magnetization. When such a point was reached, alternating field or thermal demagnetization was applied until the remanence was completely removed. One set of samples was treated initially in 10 *N* HCl, but it was found that this resulted in nearly complete removal of the NRM after 10 h and that the resolution of components was very poor.

After each treatment (thermal, chemical, alternating field) the result was plotted on an orthogonal projection (Zijderveld 1967) to determine the effect of the treatment and to estimate the next level of demagnetization that would best serve to resolve the various components. Components were identified by three or more colinear points on both projections (horizontal and vertical) of the vector diagrams. Linearity of a demagnetization trajectory on an orthogonal projection may be regarded as a determination of the spectral window of demagnetization over which a component is isolated (Dunlop 1979). Components, so determined, were isolated by vector subtraction of the two results that best defined the linear trajectory. For example, referring to Fig. 2*b*, one component is isolated by vector subtraction of the remanences measured after treatments of 550 and 640°C, and a second component is isolated by subtraction of the remanences measured after treatments of 650 and 670°C.

Components were classified on the basis of coercivity, blocking temperature, and resistance to leaching. With the exception of a single site in the Fond du Lac Formation, there was good correlation between the directions and the physical properties of the components. Components with high blocking temperatures (650–670°C) and coercivities greater than 100 mT and that resist leaching in 8 *N* HCl for up to 50–100 h were found in nearly all samples. Components with these properties were assigned to the K_1 population. A second set of components, assigned to the K_2 population, had blocking temperatures between 400 and 640°C, coercivities greater than 100 mT, and were the first components to be removed by leaching. K_2 was present in the Fond du Lac and Middle River samples, but absent from the Eileen. In almost every case, K_2 was found coexisting with K_1 in the same specimen. A third set of components, the K_3 population, was isolated only by leaching. Typically of low intensity (10–1% of the NRM), K_3 resisted up to 200 h of leaching in 10 *N* HCl. Further treatment revealed that K_3 has coercivities between 10 and 50 mT and blocking temperatures in the range of 300–400°C. When specimens were treated solely by alternating field or thermal demagnetization, K_3 was apparently removed simultaneously with K_1 or K_2.

Magnetization of the Fond du Lac Formation

Stepwise demagnetization revealed K_1, K_2, and K_3 in the NRM of samples from the fine sandstone and siltstone beds of the Fond du Lac Formation. Representative behavior of the magnetization of the samples during thermal demagnetization and leaching is presented in Fig. 2. K_1 and K_2 are clearly resolved by thermal demagnetization as evidenced by the colinear trajectories (Fig. 2 *a* and *b*). Leaching revealed K_1 and K_3 (Fig. 2 *c* and *d*) but K_2 was not well resolved. Comparison of Fig. 2*b* with Fig. 2*d* reveals that leaching led to simultaneous removal of K_1 and K_2 with a curved trajectory as the result. K_3 was revealed only after leaching and is depicted in Fig. 2*c*. Further demagnetization demonstrated K_3 was typically removed below 400°C or between 10 and 50 mT.

K_1, K_2, and K_3 were isolated by vector subtraction as previously described. The mean directions, pole positions, and statistical parameters of the three populations are given in Table 2. The Fond du Lac K_1 population is plotted in Fig. 3 *a* and *b*, and K_2 in Fig. 3 *c* and *d*. The populations isolated by thermal demagnetization are displayed in Fig. 3 *a* and *c* and by leaching in Fig. 3 *b* and *d*. Directions of opposite polarity are seen in Fig. 3*a*. Comparison of Fig. 3*a* with Fig. 3*b* shows the excellent agreement between the two methods of isolating K_1. The Fond du Lac K_2 isolated by leaching have systematically lower inclinations than do Fond du Lac K_2 isolated by thermal demagnetization. The vectors isolated by leaching represent the simultaneous removal of K_2 and the lower inclination K_1. Consequently, the K_2 population isolated thermally is interpreted to be more representative of the paleomagnetic field. Figure 4*a*

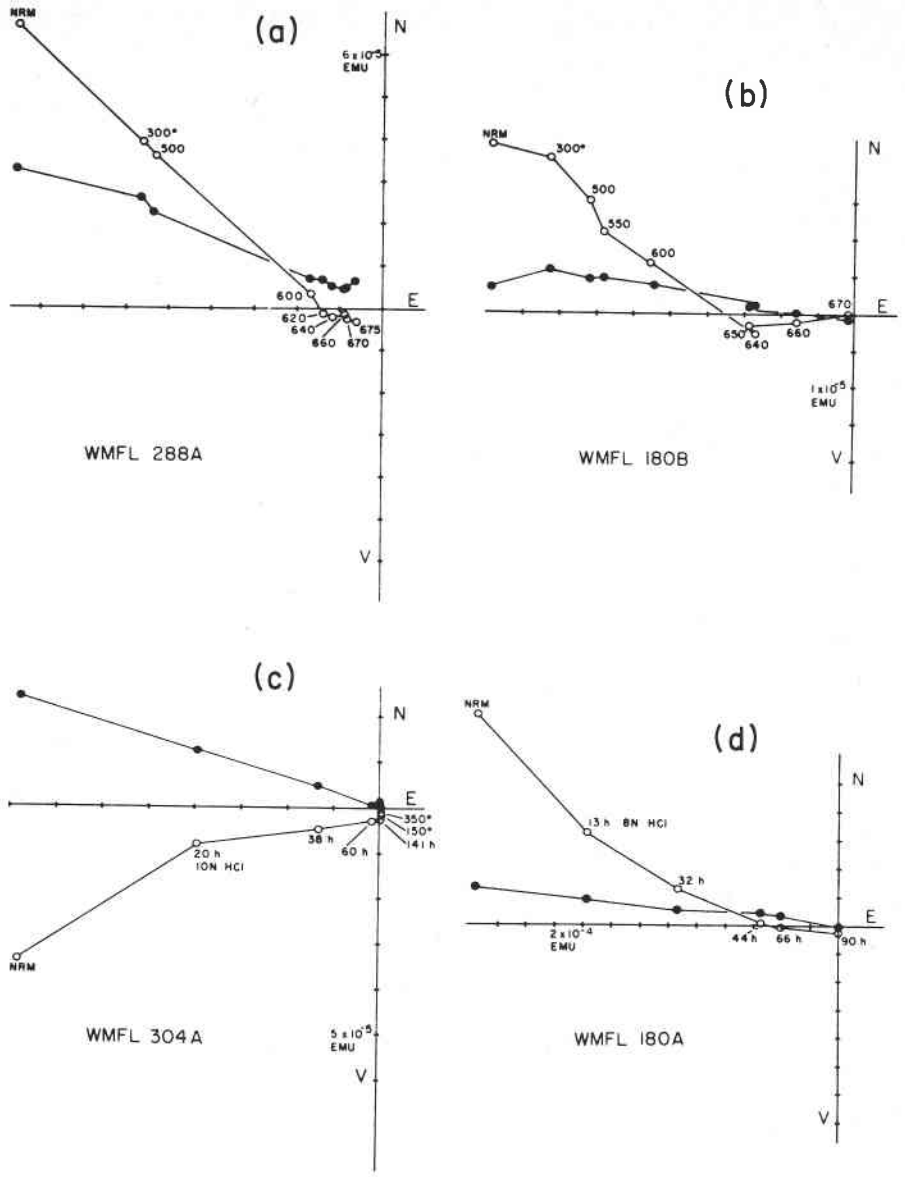


FIG. 2. Orthogonal projections (Zijderveld 1967) of demagnetization end points from representative leaching and thermal demagnetization of Fond du Lac samples. The open circles represent projections on the vertical plane, and the solid circles projections on the horizontal plane, after structural correction. K_1 and K_2 are revealed in (*a*) and (*b*). K_2 is absent from (*c*) but K_3 is revealed. The curved trajectory resulting from simultaneous removal of K_2 and K_1 by leaching is demonstrated in (*d*).

shows the K_3 population isolated from the Fond du Lac Formation. It is scattered with a mean direction near that of the present field. For reasons to be discussed, K_1 is plotted after correction for the 7° tilt of the beds, and K_2 and K_3 are plotted before correction for tilt.

The high blocking temperatures, high coercivities, and response to leaching all indicate that hematite is the carrier of K_1 and K_2. Examination of thin sections confirms Morey's (1967) report that hematite occurs as a coating on detrital grains and a filling of interstices. No hematite of detrital origin was observed and it was not possible to distinguish separate phases of hematite formation. Examination of thoroughly leached samples revealed an abundance of detrital opaque minerals. These are presumed to be magnetite or titanomagnetite because of their resistance to leaching and are the probable carriers of K_3. Although the carrier grains are detrital, it will be shown that K_3 is not.

TABLE 2. Mean directions, statistical parameters, pole positions

Population	Tilt corr.	N/R	K	A_{95}(°)	Pole Lat.(°)	Pole Long.(°)
Mean of site poles*						
Fond du Lac Formation						
K_1 thermal	Before	11/12	75	5	14	158
	After				16	160
K_2 thermal	Before	11/12	60	6	−9	145
	After				−5	148

Population	Tilt corr.	n/R_s	D(°)	I(°)	k	α_{95}(°)	Pole Lat.(°)	Pole Long.(°)	δp(°)	δm(°)
Sample mean directions										
Fond du Lac Formation										
K_1 thermal	Before	57/62	295	−6	23	4	14	157	2	4
	After		295	0			17	159		
K_1 leaching	Before	19/22	289	−2	11	11	12	163	5	11
	After		289	5			15	165		
K_2 thermal	Before	52/62	290	−48	23	4	−9	144	3	5
	After		292	−42			−4	146		
K_2 leaching	Before	17/22	295	−31	10	12	4	148	7	13
	After		296	−25			8	149		
K_3 leaching	Before	28/32	17	66	6	12	79	344	16	19
	After		30	68			70	336		
Middle River section										
K_1 thermal	Before	19/19	258	32	20	8	5	199	5	9
	After		308	−1	16	9	25	148	4	9
K_1 leaching	Before	16/17	258	32	7	15	5	199	10	17
	After		311	2	8	14	28	146	8	15
K_2 thermal	Before	19/19	83	46	46	6	−24	162	5	8
	After		50	−21	34	6	−17	212	3	6
K_2 leaching	Before	17/17	74	58	32	6	−37	158	7	9
	After		38	−28	37	6	−19	229	4	7
K_3 leaching	Before	17/17	32	75	21	8	66	308	7	9
	After		14	−36	15	10	22	74	4	7
Eileen section										
K_1 thermal	Before	16/16	276	22	14	10	12	182	6	11
	After		300	0	10	10	20	156	5	10
K_1 leaching	Before	5/5	272	26	20	18	11	187	10	19
	After		306	4	20	18	25	153	9	18
K_3 leaching	Before	5/5	323	79	119	7	61	242	13	13
	After		359	−13	75	9	37	90	5	9

NOTES: Population denotes the particular set of components used to calculate the parameters in the table; tilt corr. denotes whether the parameters are calculated before or after correction for local dip; N = the number of site poles used to calculate the mean pole; R = the number of sites analyzed; K = the estimate of Fisher's precision parameter for the dispersion of site poles; A_{95} = the radius of the cone of 95% confidence for the distribution of site poles; n = the number of samples used in the calculation of the mean directions; R_s = the number of samples analyzed; D = the mean declination; I = the mean inclination; k = the estimate of Fisher's precision parameter for the distribution of sample directions; α_{95} = the radius of the cone of 95% confidence; pole lat. = the latitude of the computed pole position; pole long. = the longitude of the computed pole position (east); δp and δm are the semiaxes of the ovals of 95% confidence around the poles computed from mean declination and inclination; see McElhinny (1973) for a complete discussion of the statistical parameters.

*A complete compilation of the Fond du Lac mean-site parameters is given in Watts (1979).

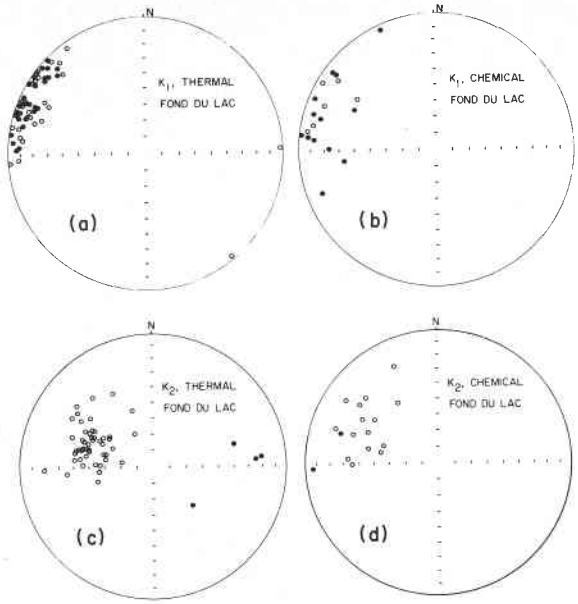


FIG. 3. Equal-area projections of the Fond du Lac K_1 and K_2. Open circles represent projections on the upper hemisphere and solid circles represent projections on the lower hemisphere. K_1 is shown isolated thermally (*a*) and by leaching (*b*) after structural correction. K_2 is shown isolated thermally (*c*) and by leaching (*d*) before structural correction.

Magnetization of the Middle River section

No systematic difference in direction or physical properties was detected between the magnetizations of the Oronto Group and the Bayfield Group in this section. As was the case for the Fond du Lac samples, the K_1, K_2, and K_3 populations were revealed by stepwise demagnetization procedures. Figure 5 *a* and *b* shows the results of typical thermal demagnetization and leaching experiments. Table 2 gives the mean direction, statistical parameters, and pole positions of the three populations, before and after structural correction.

Figure 6 *a* and *b* shows the projections of the Middle River K_1 before and after structural correction. Comparison of Fig. 6*b* with Fig. 3 *a* and *b* shows, qualitatively, the agreement with the directions of the Fond du Lac K_1 after structural correction.

Figure 6 *c* and *d* shows projections of the Middle River K_2 before and after structural correction, respectively. As was the case for the Fond du Lac, the Middle River K_2 population isolated by leaching has systematically lower inclinations compared with the K_2 isolated thermally. Again, this is attributed to simultaneous removal of K_1 and K_2. Comparing Fig. 6*c* with Fig. 3*c*, the Fond du Lac K_2 is predominantly of polarity opposite to the Middle River K_2 before structural correction. An argument will be presented favouring a post-tectonic age for the Middle River K_2.

The Middle River K_3 population is plotted without structural correction in Fig. 4*b*. This population clusters about the present field direction; structural correction carries the mean to a direction similar to certain North American Ordovician magnetizations. However, this is not consistent with a pre-Late Cambrian age of deformation in the Keweenawan Basin. It is concluded that the Middle River K_3 is post-tectonic.

Thin-section examination shows an abundance of detrital opaque grains remaining after chemical leaching, as was the case for the Fond du Lac Formation. Since a post-tectonic age of magnetization is not consistent with a detrital magnetization and because of the comparatively low blocking temperatures and coercivities of the Fond du Lac and Middle River K_3 populations, K_3 is interpreted to be a very recent viscous magnetization acquired in the present field.

Magnetization of the Eileen section

The component structure of the NRM of the Eileen section is significantly different from those of the Fond du Lac Formation and the Middle River section. Examination of the orthogonal projections of the nearly univector trajectories in Fig. 7 shows that K_2 is not recorded in the Eileen section. Low intensity secondary

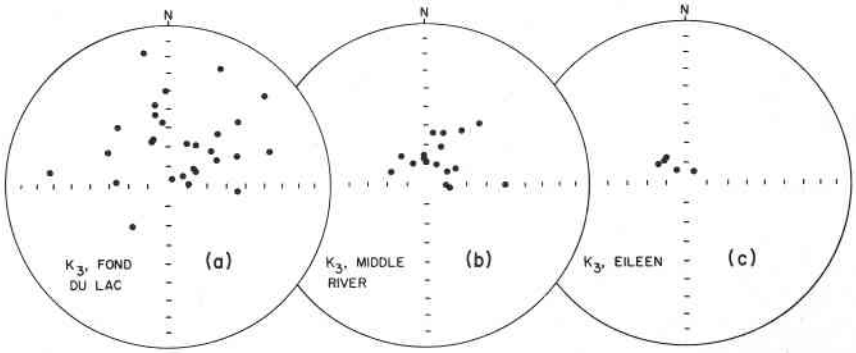


FIG. 4. Equal-area projections of K_3 from the Fond du Lac (*a*), Middle River (*b*), and Eileen (*c*) sections before structural correction.

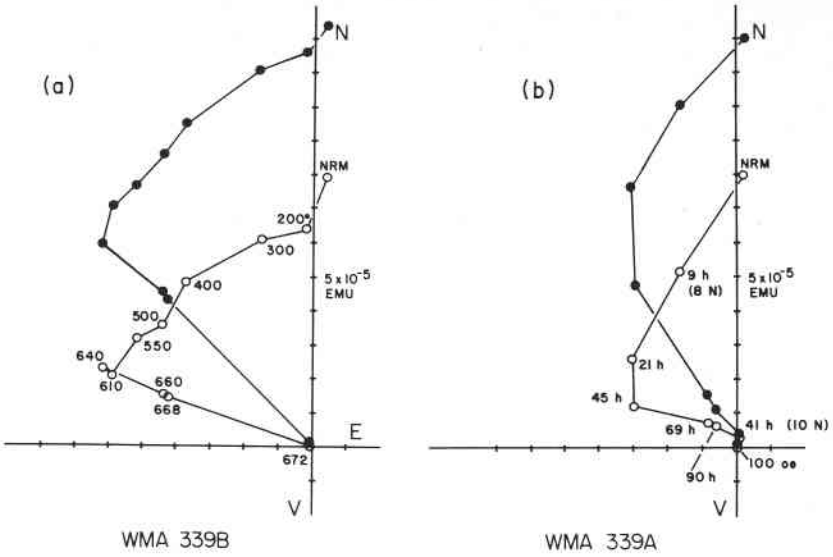


FIG. 5. Orthogonal projections of results from Middle River section samples showing thermal demagnetization (*a*) and leaching (*b*). K_1 and K_2 are revealed in both diagrams. K_3 is depicted in (*b*). The results are plotted after structural correction.

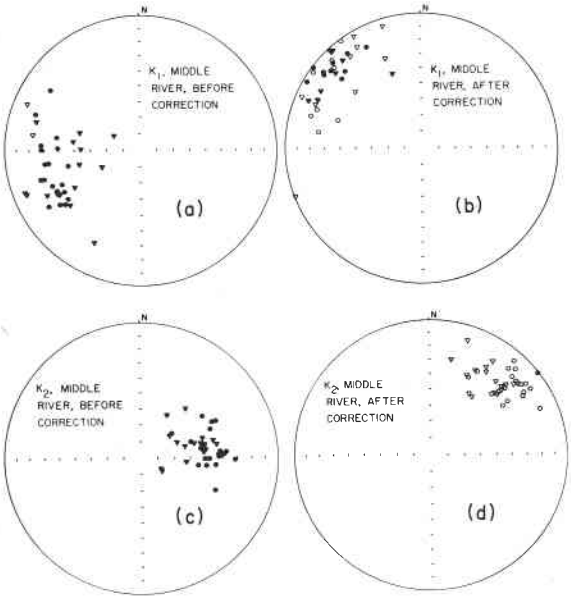


FIG. 6. Equal-area projections of the Middle River K_1 and K_2. Circles represent results from thermal demagnetization, triangles from leaching. K_1 is plotted before and after structural correction in (*a*) and (*b*), respectively. K_2 is plotted before and after structural correction in (*c*) and (*d*), respectively.

components are present at the low blocking temperature range of some samples and may possibly be assigned to K_3 but these are not well resolved by thermal demagnetization. K_1 dominates the NRM of these samples, and is well resolved by thermal demagnetization and leaching. K_3 is clearly revealed by leaching and is shown to have low blocking temperatures (350°C) by subsequent thermal demagnetization (Fig. 7*a*).

Plots of the Eileen K_1 are given in Fig. 8 *a* and *b*, before and after structural correction. Qualitative comparison of Figs. 3*a*, 6*b*, and 8*b* shows the similarity of the directions of K_1 from the three sections after structural correction. Because of the nearly univector nature of magnetization of the Eileen samples, the mean K_1 direction, isolated thermally (Table 2), is similar to a mean paleomagnetic direction derived by DuBois (1962) from limited laboratory treatment.

The Eileen K_3, plotted in Fig. 4*c*, is near the present field before structural correction as are the Fond du Lac K_3 and the Middle River K_3. The Eileen K_3 is interpreted to be a very recent viscous magnetization.

Pole positions

Pole positions for the Fond du Lac Formation, the Middle River section, and the Eileen section were calculated from the K_1 populations isolated by thermal demagnetization. The Fond du Lac pole is a mean calculated from the site poles. The details of the mean-site directions, poles, and statistics are given in Watts (1979). Before structural correction, the pole is 14°N, 158°E, and after correction, 16°N, 160°E. The Middle River and Eileen poles are sample-mean poles. The Middle River pole lies at 5°N, 199°E before correction and 25°N, 148°E after correction. The Eileen K_1 pole lies at 12°N, 182°E before correction and 20°N, 156°E after. Figure 9 shows the poles before and after structural correction. The similarity of the K_1 directions between the three sections after tilt correction may be demonstrated in a quantitative manner by the calculation of the dispersion of the associated poles before and after cor-

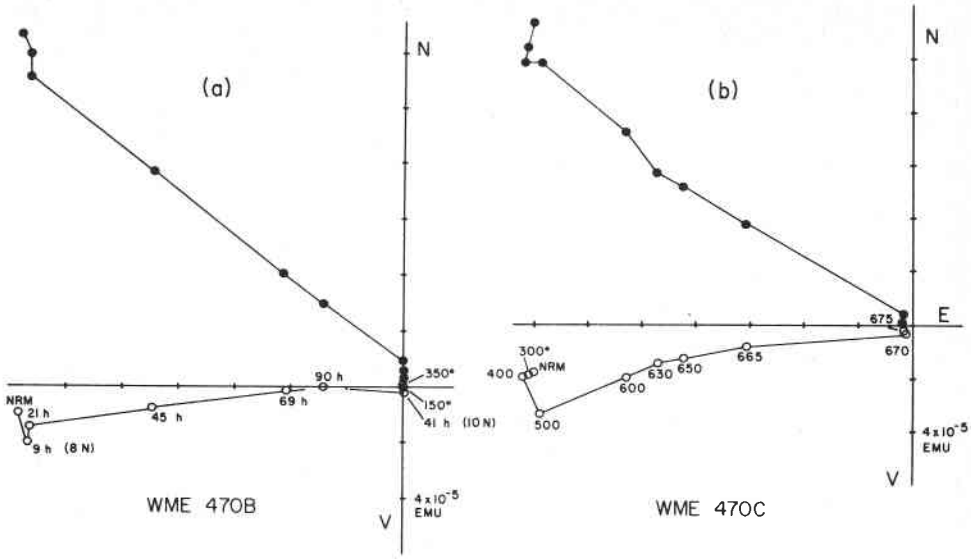


FIG. 7. Orthogonal projections of leaching (*a*) and thermal demagnetization (*b*) results of Eileen section samples. Note the dominance of K_1 and the absence of K_2. K_3 is well resolved in (*a*).

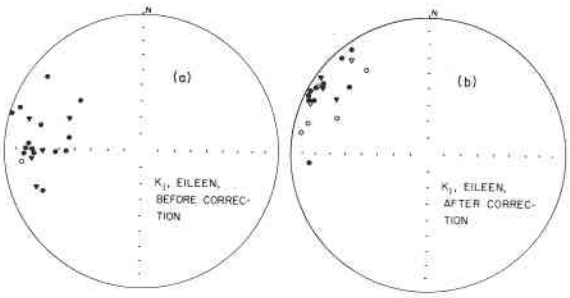


FIG. 8. Equal-area projections of Eileen K_1 before (*a*) and after (*b*) structural correction. Circles represent results from thermal demagnetization and triangles from leaching.

rection. Before correction, the poles form a group with a mean of 6°N, 183°E, $K = 12$, $A_{95} = 37°$, and after correction: 20°N, 155°E, $K = 124$, $A_{95} = 11°$. The clustering is significant to the 95% level of confidence when the test of Watson (1956) using the tables calculated by McElhinny (1964) is applied to the data. The clustering of the poles is regarded as evidence of the pre-tectonic age of K_1.

Petrographic observations, as previously discussed, show hematite as a coating on detrital grains and as filling interstices. K_1 may not be of detrital but of chemical origin, acquired some time after deposition but before lithification. The Fond du Lac Formation, Middle River section, and the Eileen section acquired K_1 in the same paleomagnetic field, before the deformation of the Keweenawan Basin, but this does not necessarily imply that the deposition of the three sections was simultaneous.

The K_2 poles were calculated only from thermal demagnetization results because of simultaneous removal of components by leaching. The Fond du Lac K_2 pole, a mean of site poles, falls at 9°S, 145°E before structural correction and 5°S, 148°E after. The Middle River K_2 pole falls at 24°S, 162°E before structural correction and 17°S, 212°W after. Examination of Fig. 9 shows that these poles diverge after structural correction. Also, the Middle River K_2 pole, after structural correction, does not lie near any previously published North American poles younger than ca. 1100 Ma. These observations are interpreted to demonstrate Middle River K_2 was acquired after the tilting of the Middle River section. The Fond du Lac K_2 is therefore interpreted to be younger than the Fond du Lac K_1 and to postdate the slight tilting of the Fond du Lac Formation. The Fond du Lac K_2 pole is statistically distinguishable from the Middle River K_2 pole, separated by 20° of arc. The Fond du Lac K_2 population is almost entirely of opposite polarity compared with the Middle River K_2. This may indicate the K_2 populations from the respective sections are of different ages.

The origin of K_2 is a problem. With blocking temperatures up to 640°C, a thermal remanent origin appears unlikely. Theoretical time–temperature curves of the unblocking of hematite (Pullaiah *et al.* 1975) show that temperatures of the order of 600°C are required for thermal resetting of hematite grains with such blocking temperatures, no matter how long the rocks are heated. Such high temperatures are inconsistent with the unmetamorphosed character of the host rock. Because the K_2 population postdates the deformation of the Keweenawan Basin, it was acquired after lithification. The K_2 populations may be related to a chemical altera-

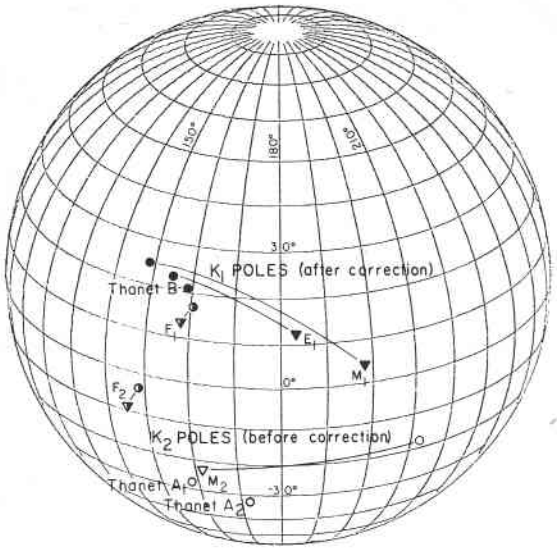


FIG. 9. Pole positions calculated from K_1 and K_2 showing poles before (triangles) and after (circles) structural correction. The open characters denote poles of polarity opposite to the poles denoted by closed characters. The partially open characters denote poles of mixed polarity. F_1 is calculated from the Fond du Lac K_1, M_1 from the Middle River K_1, F_2 from the Fond du Lac K_2, M_2 from the Middle River K_2. The clustering of the K_1 poles and the divergence of the K_2 poles after structural correction are shown. The Thanet gabbro B, A_1, and A_2 poles are plotted for comparison with the Middle River (M_1 and M_2) poles. The similarity of the magnetic signature of the Thanet gabbro and the Middle River section is clearly demonstrated.

tion induced during the burial of these rocks during deformation.

The path of apparent polar wander for North America

A number of interpretations of the geometry of the path of apparent polar wander of North America for the period 1100–650 Ma have been published (e.g., Irving *et al.* 1974; Ueno *et al.* 1975; Irving and McGlynn 1976; Morris and Roy 1977; Roy and Robertson 1978; McWilliams and Dunlop 1978; Berger *et al.* 1979). As discussed in these studies, a major problem in North American paleomagnetism is the interpretation of paleomagnetic poles from the Grenville Province. A further problem is the difficulty of obtaining ages of magnetization from the ages of closure of isotopic systems (Buchan *et al.* 1977) in igneous and metamorphic terrains. The uncertainties inherent in isotopic ages and their geologic significance compound the difficulty of assigning the appropriate relative ages of magnetic components found in metamorphic terrains. The thermochron approach of McWilliams and Dunlop (1978) and the calibration of paleopoles by $^{40}Ar/^{39}Ar$ dating (Berger *et al.* 1979) are contributions toward the solution of this problem.

Although is it not possible to assign absolute ages to the magnetization of the Fond du Lac Formation and the Middle River and Eileen sections, the relative ages of magnetization are determined from the structural relationships. Any interpretation of the geometry of the apparent path of polar wander for North America must therefore accommodate the sequence of magnetization $K_1 \rightarrow K_2$. The K_2 poles reported in this study fall among the Grenville poles (Fig. 10) and support those interpretations that include the Grenville poles as an integral part of the path of apparent polar wander of Laurentia.

A magnetic component structure similar to that of the Middle River section has been reported from the Thanet gabbro of the Grenville Province (Buchan 1978). The Thanet B pole, at 20°N, 159°E, falls among the K_1 poles and the Thanet A_1 pole, at 28°S, 158°E, falls among the Grenville poles and is 5° of arc away from the Middle River K_2 pole. The sequence of magnetization $B \rightarrow A$ is consistent with a simple cooling model of the Thanet gabbro and with the order of magnetization of the Middle River section.

The thermochron interpretation of the Grenville poles (McWilliams and Dunlop 1978) is consistent with a sequence of magnetization inferred from the stratigraphic order of the Keweenawan sediments (Table 1) and the sequence of magnetization inferred from the structural relations. The sequence of magnetization $K_1 \rightarrow K_2$ is also consistent with the path of Berger *et al.* (1979), which is sketched in Fig. 11. This track is partially calibrated by $^{40}Ar/^{39}Ar$ age determinations,

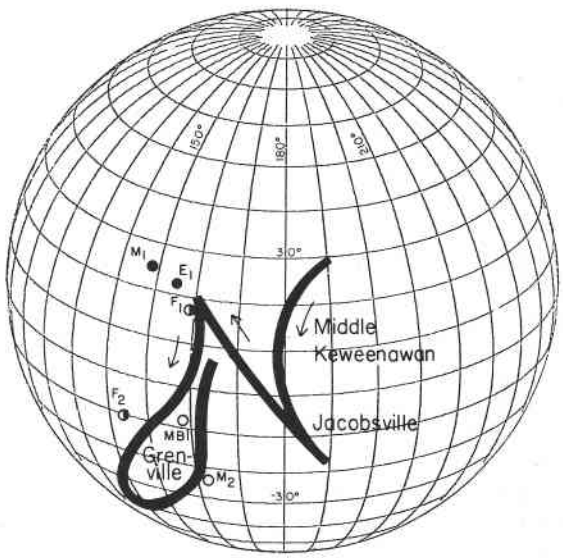


FIG. 10. The path of apparent polar wander after McWilliams and Dunlop (1978) with K_1 poles and K_2 poles labeled as in Fig. 9. MBI is the Michigan borehole igneous rock pole.

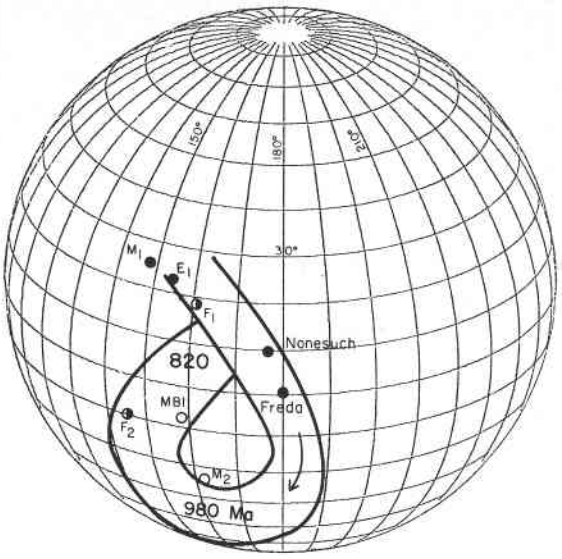


FIG. 11. The path of apparent polar wander after Berger *et al.* (1979) with Upper Keweenawan and Michigan borehole igneous rock poles. The characters are as described in Fig. 9.

which are given in the figure. This interpretation does not follow the sequence of magnetization given by the stratigraphic order of the Keweenawan sediments. The Eileen, Middle River, and Fond du Lac K_1 poles appear to be older than the Freda and Nonesuch poles, which is not the sequence given in Table 1. The K_1 poles cannot lie on the youngest part of the proposed loop because this would violate the sequence $K_1 \rightarrow K_2$. It must be emphasized that the stratigraphic sequence may not reflect the order of magnetization if the magnetizations are chemical rather than detrital.

Another difficulty in the interpretation is the polarity problem: are the poles south poles or north poles? If the Middle River K_2 pole is plotted in the western hemisphere, it falls on the Hadrynian track of Morris and Roy (1977). This implies an age of K_2 much younger than Grenvillian. It is not possible to establish absolute ages of magnetization in Precambrian sedimentary rocks; only relative ages may be inferred. However, because of the similarity of the magnetic signature of the Middle River section to the magnetic signature recorded in the Thanet gabbro, even to the polarity of the components, a Grenville age of K_2 is favoured.

The K_3 population gives poles that fall on the Hadrynian track of Morris and Roy (1977) and components with a similar direction and polarity are found in the Jacobsville sandstone (the J_2 population of Roy and Robertson 1978). The blocking temperatures and coercivities of the Jacobsville J_2 are substantially higher than those observed in the K_3 populations. It is previously argued that the steeply inclined magnetic components reported in this paper are of recent origin. Because of the substantially different physical properties, this does not preclude a Precambrian age for the Jacobsville J_2.

Van der Voo and Watts (1978) reported paleomagnetic results from a basic igneous rock unit and overlying red beds from a deep borehole drilled near the center of the Michigan Basin. The igneous rock pole is plotted in Figs. 10 and 11. Their tentative conclusion that the igneous rock was remagnetized in Late Hadrynian or Cambrian time is now called into question because the igneous rock pole falls directly on the Grenville track.

Keweenawan tectonism

From the above interpretations, a number of conclusions may be drawn about the nature and timing of tectonism in the Keweenawan Basin. The outpouring of enormous volumes of Keweenawan basalt records a phase of rifting or a tensional regime, with a pole of spreading for the aborted mid-continental rift as calculated by Chase and Gilmer (1973). McWilliams and Dunlop (1978) suggest a casual relationship between events in the Grenville Province and the rifting phase in the Keweenawan Basin. Rapid accumulation of the terrestrial Upper Keweenawan sediments followed the rifting. Marine sediments are absent from surface outcrops, but Fowler and Kuenzi (1978) report the occurrence of marine turbidite sequences from the subsurface of Michigan. This was followed by a compressive regime during which movement along the Douglas, Keweenawan, and other thrust faults was initiated, accompanied by the formation of the Lake Superior syncline and other folds in the Keweenawan Peninsula of Michigan and northern Wisconsin. The age of this compressive event is bracketed by the times of acquisition of K_1 and K_2.

The K_1 poles are not accurately dated but they must postdate the Keweenawan rhyolite flows, which underlie the Oronto Group in the Keweenawan Peninsula of Michigan. These flows have Rb/Sr dates ranging from 1088 to 1040 Ma (Chaudhuri, quoted in Henry *et al.* 1977). The age of K_2 is problematical because of uncertainties in the calibration of the Grenville track. The calibration age of Berger *et al.* (1979) of 820 Ma is based on the closure of biotite K/Ar systems. This age may be too young for the time of magnetization of single-domain magnetite grains, which block at substantially higher temperatures than does argon in biotite. Morris and Roy (1977) give a minimum age for the Grenville magnetizations of 950 Ma. The dating of tectonism in the Keweenawan Basin awaits the precise calibration of the Grenville polar wander track, but at this point, accepting a minimum age of 950 Ma for Grenville magnetizations, it is estimated that the compressive event in the Keweenawan Basin is bracketed between 950 and 1040 Ma. If this interpretation is valid, the timing is such that the compressive event in

the Keweenawan Basin may be related to the Grenville Orogeny.

Conclusions

Thermal and alternating field demagnetization and leaching of oriented samples from the Fond du Lac Formation and the Middle River and Eileen sections of the Upper Keweenawan sediments reveal three components of the NRM. These components may be classified by the physical properties of coercivity, blocking temperature, and resistance to leaching.

The K_1 population is found in all sections sampled in this study. It is characterized by a high blocking temperature (650–670°C), coercivity greater than 100 mT, and elimination after 60–100 h of leaching in 8 *N* HCl. K_1 is pre-tectonic, acquired before the tilting of the Middle River and Eileen sections.

The K_2 population is characterized by a coercivity greater than 100 mT, blocking temperatures between 400 and 640°C, and elimination of the carrier before 60 h of leaching in 8 *N* HCl. K_2 is found in the Fond du Lac Formation and the Middle River section, but is absent from the Eileen. It is post-tectonic, acquired after the tilting of the Middle River section.

The K_3 population was resolved only by leaching. It is characterized by blocking temperatures between 300 and 400°C and coercivities from 10 to 50 mT. It is found in all three sections and is post-tectonic. The coercivity and blocking temperature suggest magnetite or titanomagnetite as the carrier of the K_3 population. Because of its comparitively low coercivity, low blocking temperature, nondetrital origin, and proximity to the present field, K_3 is considered to be a very recent viscous magnetization.

Any interpretation of the path of apparent polar wander with respect to North America must account for the sequence of magnetization $K_1 \rightarrow K_2$. Any such interpretation must accommodate the K_2 poles that fall among the Grenville poles. Two recently published interpretations that satisfy both of these requirements are those of McWilliams and Dunlop (1978) and Berger *et al.* (1979).

The age of deformation in the Keweenawan basin is bracketed by the times of acquisition of K_1 and K_2. K_1 must postdate the Middle Keweenawan rhyolitic lavas that have Rb/Sr dates between 1088 and 1040 Ma. The minimum age of Grenville magnetizations is given by Morris and Roy (1977) as 950 Ma. Accepting the premise that K_2 is of Grenville age, the age of the compressive event in the Keweenawan Basin is bracketed between 1040 and 950 Ma.

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